

Observation Task-4

1. Basic Information

Date: 21/09/2026

Title: Student Performance Analysis System

2. Objective / Aim

To develop a simple Java program using **classes, objects, constructors, arrays, methods, String operations, and Math methods** to analyze student performance.

3. System Requirements

Hardware Requirements:

- Computer/Laptop
- Minimum 4 GB RAM

Software Requirements:

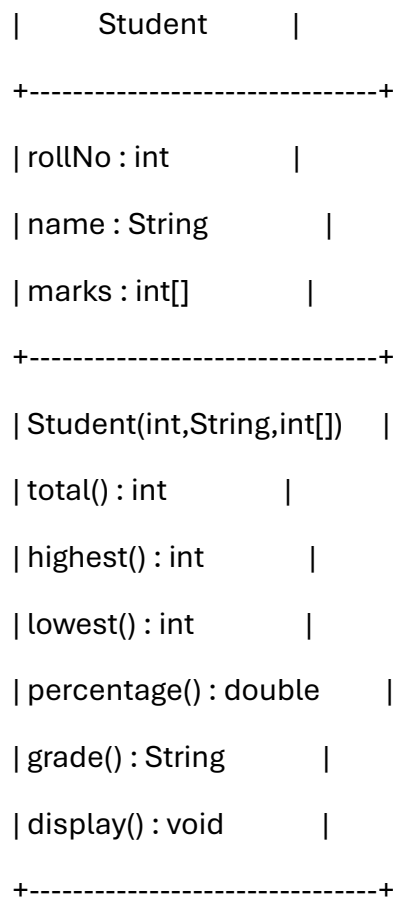
- Windows / Linux / macOS
- JDK 17 or JDK 21
- Eclipse / IntelliJ IDEA / NetBeans / VS Code

4. Algorithm / Logic Description

1. Create a Student class with roll number, name, and marks array.
2. Create a constructor to initialize student details.
3. Store marks of five subjects in an array.
4. Calculate total and average marks using a loop.
5. Find highest and lowest marks using Math.max() and Math.min().
6. Calculate and round the percentage using Math.round().
7. Process the name using trim(), toUpperCase(), and length().
8. Assign a grade based on the percentage.
9. Display PASS if percentage is 50% or above; otherwise display FAIL.
10. Display the performance remark and complete student report.

5. UML Class Diagram

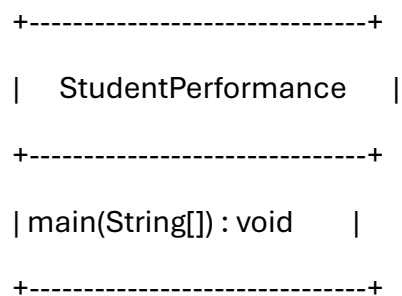
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6. Source Code

```

class Student {
    int rollNo;
    String name;

```

```
int[] marks;
```

```
Student(int rollNo, String name, int[] marks) {  
    this.rollNo = rollNo;  
    this.name = name;  
    this.marks = marks;  
}
```

```
int total() {  
    int t = 0;  
    for (int m : marks)  
        t += m;  
    return t;  
}
```

```
int highest() {  
    int h = marks[0];  
    for (int m : marks)  
        h = Math.max(h, m);  
    return h;  
}
```

```
int lowest() {  
    int l = marks[0];  
    for (int m : marks)  
        l = Math.min(l, m);  
    return l;  
}
```

```
double percentage() {  
    return Math.round(total() / 5.0 * 100) / 100.0;  
}
```

```
String grade() {  
    double p = percentage();  
  
    if (p >= 90) return "A+";  
    if (p >= 80) return "A";  
    if (p >= 70) return "B";  
    if (p >= 60) return "C";  
    if (p >= 50) return "D";  
    return "F";  
}
```

```
void display() {  
    String n = name.trim().toUpperCase();  
  
    System.out.println("STUDENT PERFORMANCE REPORT");  
    System.out.println("Roll No    : " + rollNo);  
    System.out.println("Name      : " + n);  
    System.out.println("Name Length : " + n.length());  
    System.out.println("Total     : " + total() + "/500");  
    System.out.println("Average   : " + total() / 5.0);  
    System.out.println("Highest   : " + highest());  
    System.out.println("Lowest    : " + lowest());  
    System.out.println("Percentage : " + percentage() + "%");  
}
```

```

        System.out.println("Grade    : " + grade());
        System.out.println("Result   : " + (percentage() >= 50 ? "PASS" : "FAIL"));
        System.out.println("Remark   : " +
            (grade().equals("A+") ? "Excellent" :
            grade().equals("A") ? "Very Good" :
            grade().equals("B") ? "Good" :
            grade().equals("C") ? "Satisfactory" :
            grade().equals("D") ? "Needs Improvement" : "Work Hard"));
    }
}

```

```

public class StudentPerformance {
    public static void main(String[] args) {
        int[] marks = {85, 92, 78, 88, 95};

        Student s = new Student(101, " Jaswanth ", marks);
        s.display();
    }
}

```

7. Compilation & Execution Commands

File Name:

StudentPerformance.java

Compile:

javac StudentPerformance.java

Run:

java StudentPerformance

8. Output Screen

STUDENT PERFORMANCE REPORT

Roll No : 101

Name : JASWANTH

Name Length : 8

Total : 438/500

Average : 87.6

Highest : 95

Lowest : 78

Percentage : 87.6%

Grade : A

Result : PASS

Remark : Very Good

9. Result

The **Student Performance Analysis System** was successfully implemented in Java using **class, object, constructor, array, methods, loops, String operations, Math methods, and conditional statements**. The program successfully analyzes and displays the student's academic performance.